Supplementary Materials: Scoring Rubric and Retrieval Diagnostics for Dual-Source RAG

This supplement reports the answer-quality rubric and the candidate-budget replay and BM25 locator-granularity comparisons cited in the main text. The rubric was developed after reviewing output availability and some answers; it is an exploratory descriptive scale, not a preregistered H1/H2 test. Retrieval diagnostics reuse saved rankings or execute retrieval; they do not call a generation model, add answer ratings, or add human relevance labels. These results describe evidence supply and locator boundaries; they do not establish answer accuracy, retrieval precision/recall, or historical regulatory applicability. Reference numbering follows the main text.

Supplementary Material S0. Answer-Quality Scoring Rubric

Two raters independently score each actual answer on four dimensions using 0, 1, 2, or U. The anchors below reproduce the original scoring table; existing ratings are not reassigned.

Dimension	Score 0	Score 1	Score 2
Company facts	A key company, number, member, date, or reporting-period fact is wrong or fabricated; or the answer refuses without reason when the context is sufficient.	Facts are generally reliable, but key points are omitted or incomplete. Missing input is not treated as proof that a company event did not occur.	Accurately covers the main facts and distinguishes reporting-period from subsequent events. If the source evidence is insufficient, clearly states the gap without fabrication; this may also receive 2.
Regulatory basis	Misquotes or misinterprets a key rule, or substitutes company disclosure for a requested regulatory basis without acknowledging the gap.	Cites a relevant rule but omits a key requirement or gives incomplete scope/version caveats.	Accurately states the main requirements and links them to the source text. States applicability when verifiable and qualifies it when unknown; other valid sources may be used if checked against the original.
Citation support	A key citation is absent, contradictory, or misattributed, or substantive content has no support. A valid citation ID alone earns no credit.	Some citations support the answer, but important claims lack citations, source roles are omitted, or locators are imprecise; support is insufficient overall.	Substantive claims are supported by the cited original text; annual-report and regulatory sources are cited as needed, with traceable sources and page/article locators. Reasonable caveats when evidence is insufficient are not penalized.
Uncertainty handling	Treats self-evaluation as external assurance, a financial-statement audit as an internal-control audit, or a proposal as implemented; or asserts absence, compliance, or violation without support.	Includes caveats but omits an important boundary or limitation; no decisive unsupported certainty is present.	Follows the final adjudication limits; distinguishes self-assessment/audit, implementation/proposal, and fact/rule; does not infer absence or full compliance from unknown information.

Scoring and unknown values. U means that a rater cannot determine the score and must state why; it is not a score of zero. A 0–8 answer score is computed as the mean of the two raters' four-dimension totals only when both raters provide numeric scores on all four dimensions. Failed or unrun requests are not assigned a quality score of zero. This answer rating is not a governance score, compliance accuracy rate, or retrieval relevance metric.

Supplementary Material S1. Budget-Quota Replay and Evidence Supply

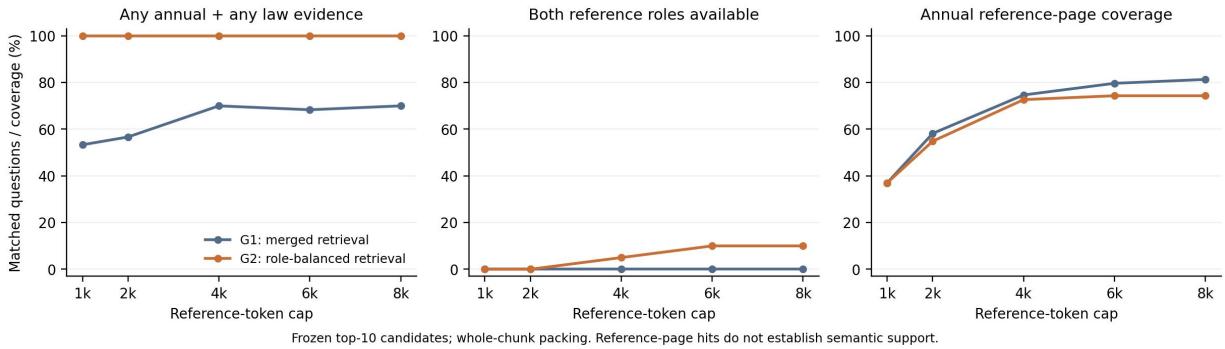
The 1,800 contexts span five token caps and three candidate quotas, using the same candidate order as the historical runs. All 120/120 contexts under the original configuration were reproduced character for character, showing that subsequent changes came from the specified candidate truncation and packing, not from a different implementation. A larger budget can admit more complete passages, but cannot recover a target source passage that is absent from the saved candidate pool.

Under the original 6,000-token configuration, all ten G2 candidates were packed, yet the final context still lacked a regulatory target location for 53/60 questions. G1 did not strictly hit the regulatory reference location for any of the 60 questions. Thus, for these target source–page keys, the main difference arose during candidate selection rather than default G2 token packing. Equivalent text retrieved from another source would not count toward this measure and would require independent semantic alignment.

This experiment does not search outside the candidate pool or generate 1,800 answers. It tests budget sensitivity for the same saved rankings; it does not establish the end-to-end effect of rerunning neural retrieval or reranking with a different top-k. Any inference about quality from this high-budget replay must remain at the evidence-supply level; answer-quality gains require actual generation and independent ratings. Results appear in Supplementary Table S1 and Supplementary Figure S1.

Supplementary Table S1. Budget Replay (Maximum of 10 Total Candidates)

Strategy	Token cap	Passages retained	Actual tokens	Both evidence roles present	Macro-average locator coverage by role
dual	1000	3.77	953	100.0%	18.5%
dual	2000	6.18	1905	100.0%	27.4%
dual	4000	9.10	3448	100.0%	37.4%
dual	6000	10.00	4032	100.0%	39.1%
dual	8000	10.00	4032	100.0%	39.1%
merged	1000	2.73	868	53.3%	18.5%
merged	2000	4.47	1875	56.7%	29.1%
merged	4000	7.55	3726	70.0%	37.3%
merged	6000	9.37	4945	68.3%	39.8%
merged	8000	10.00	5407	70.0%	40.7%



Supplementary Figure S1. Additional budget sensitivity with a maximum of 10 total candidates. Curves show replay of cached candidates and exclude new retrieval, generation, and human ratings.

Supplementary Material S2. BM25 Baseline and Locator-Granularity Sensitivity

To avoid merely redrawing existing results, we ran a new CPU BM25 retrieval over the same scope. We used the project's original BM25 and jieba implementation (k1 = 1.5, b = 0.75); IDF was fitted over all 19,367 passages, matching the original hybrid-retrieval component. Retrieved candidates still excluded 335 obsolete-version passages and were selected only from the 19,032 eligible passages, with annual reports restricted by company and year. Public questions were used directly as queries, and ranking IDs were saved before reference answers were read to prevent constructing favorable queries from reference citations.

All 60 questions were run under both merged and dual retrieval, and each ranking was packed under five budgets, producing 600 new contexts. This is a newly executed sparse-retrieval run, not a repacking of cached neural rankings. No generator was called; therefore, it cannot replace the previously human-rated BM25 generation condition, and the two BM25 conditions must not be combined as one end-to-end experiment. Source-location results for rankings and budget replay appear in Supplementary Table S2.

Supplementary Table S2. New BM25 versus Cached Hybrid Reranking at a 6,000-Token Cap: Source-Location Proxies

Retrieval source	Strategy	Annual-report locator coverage	Regulatory locator coverage	Both reference roles	Actual tokens
Cached hybrid reranking	dual	74.3%	3.9%	10.0%	4032
Cached hybrid reranking	merged	79.7%	0.0%	0.0%	4945
New BM25	dual	69.0%	33.3%	33.3%	5319
New BM25	merged	73.0%	30.0%	28.3%	5381

At first glance, the new BM25 run has higher regulatory-location coverage (30.00% and 33.33%) than cached hybrid reranking. The granularity breakdown shows that this increase comes entirely from matching whole TXT sources: for the 20 internal-control questions, merged and dual BM25 hit 18/20 and 20/20 sources, respectively. For the 40 questions with PDF regulatory references, both BM25 strategies have 0 page-level coverage, while cached dual retrieval reaches 5.83%. The aggregate increase therefore mainly reflects locator granularity and document identity; it does not show that BM25 retrieves more precise legal passages.

This sensitivity analysis shows that page-level and source-file-level labels in mixed-document RAG evaluation should not be combined into a single "retrieval accuracy" measure without explanation. Retrieving the full regulation file is better than not retrieving it, but does not guarantee that the returned passage contains the relevant provision or the applicable version. Results should remain separated when their units differ, so a system

is not favored merely because it matches a coarser label. PDF page-level and TXT file-level results are separated in Supplementary Table S3.

Supplementary Table S3. Regulatory Locator Granularity (PDF Page Level: 40 Questions, 120 Target Locations; TXT Source Level: 20 Questions; Denominators Differ and Are Not Directly Comparable)

Method	Strategy	PDF-page coverage (120 target locations)	TXT-source hits (20 questions)
Cached hybrid reranking	merged	0.00%	0 / 20
Cached hybrid reranking	dual	5.83%	0 / 20
New BM25	merged	0.00%	18 / 20
New BM25	dual	0.00%	20 / 20

As an alternative-source sensitivity check, 12-character lexical coverage of regulatory reference citations without restricting source identity is 0.59% for cached G1 and 11.88% for cached G2. For 33 questions, G2 did not hit the strict reference location but some reference wording appeared on another source or page. Possible reasons include restated rules, amendment decisions, or duplicate text in the body of the regulation. Lexical overlap does not establish regulatory equivalence, version identity, or support for a specific claim. It only suggests that strict locators may miss alternative material; the two measures must not be added into a semantic accuracy score.

This comparison remains subject to component-scope limitations: cached hybrid results use previously saved neural candidates, whereas new BM25 reruns sparse retrieval; the full neural pipeline was not rerun alongside it. Model revisions and original queries are recorded for future reproduction on the same machine and within a single execution. The purpose is to clarify the relationship among metrics, locator granularity, and evidence supply, not to claim that one complete system wins.